Year II — Mistake & Methods Log

Based on lectures by Various Lecturers

Notes taken by Anonymous

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Chapter 1

Mathematical Methods

1.1 Derivatives and Coordinates

1.1.1 Mistakes

First mistake

1.1.2 Methods

Method 1.1.1 (Evaluating a Scalar Line Integral)**.**

(i) Write $f(\mathbf{x}), \mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))$ and $t \in [t_0, t_1]$

(ii) Find $\frac{d\mathbf{x}}{dt}$ and then take the magnitude (this may not be an integer but a function of t)

(iii) Substitute in and integrate

Remark 1.1.2. You may need to parametrise first if given the implicit formula. Make sure to remember conversions to polar coordinates for circles. Sketching is therefore also useful.

Method 1.1.3 (Evaluating arc length)**.**

(i) Write $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))$

(ii) Find $\frac{d\mathbf{x}}{dt}$

(iii) Perform $\sqrt{\frac{d\mathbf{x}}{dt} \cdot \frac{d\mathbf{x}}{dt}}$

(iv) Use the formula for arclength

Method 1.1.4 (Evaluating a Vector Field Line Integral)**.**

(i) Sketch $\mathbf{x}(t)$ if possible

(ii) Substitute $\mathbf{x}(t)$ into $\mathbf{F}(\mathbf{x})$ to form $\mathbf{F}(\mathbf{x}(t))$

(iii) Find $\frac{d\mathbf{x}}{dt}$

(iv) Perform $\mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt}$

(v) Substitute into the formula

Remark 1.1.5. When $\mathbf{x}(t)$ is defined piecewise, makesure to split the integral.

Method 1.1.6 (Finding unit tangent vectors)**.** To find the unit tangent vectors for the coordinate system $\mathbf{x}(u, v, w)$ use the formulas:

$$\boldsymbol{e}_u = \frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}, \quad \boldsymbol{e}_v = \frac{1}{h_v}\frac{\partial \boldsymbol{x}}{\partial v}, \quad \boldsymbol{e}_w = \frac{1}{h_w}\frac{\partial \boldsymbol{x}}{\partial w},$$

where

$$h_u = \left|\frac{\partial \boldsymbol{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \boldsymbol{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \boldsymbol{x}}{\partial w}\right|$$

are called the scale factors.

Method 1.1.7 (Ensuring an Orthonormal Basis)**.** To check if these unit tangent vectors form an orthonormal basis we will have to ensure that they are all mutually orthogonal.